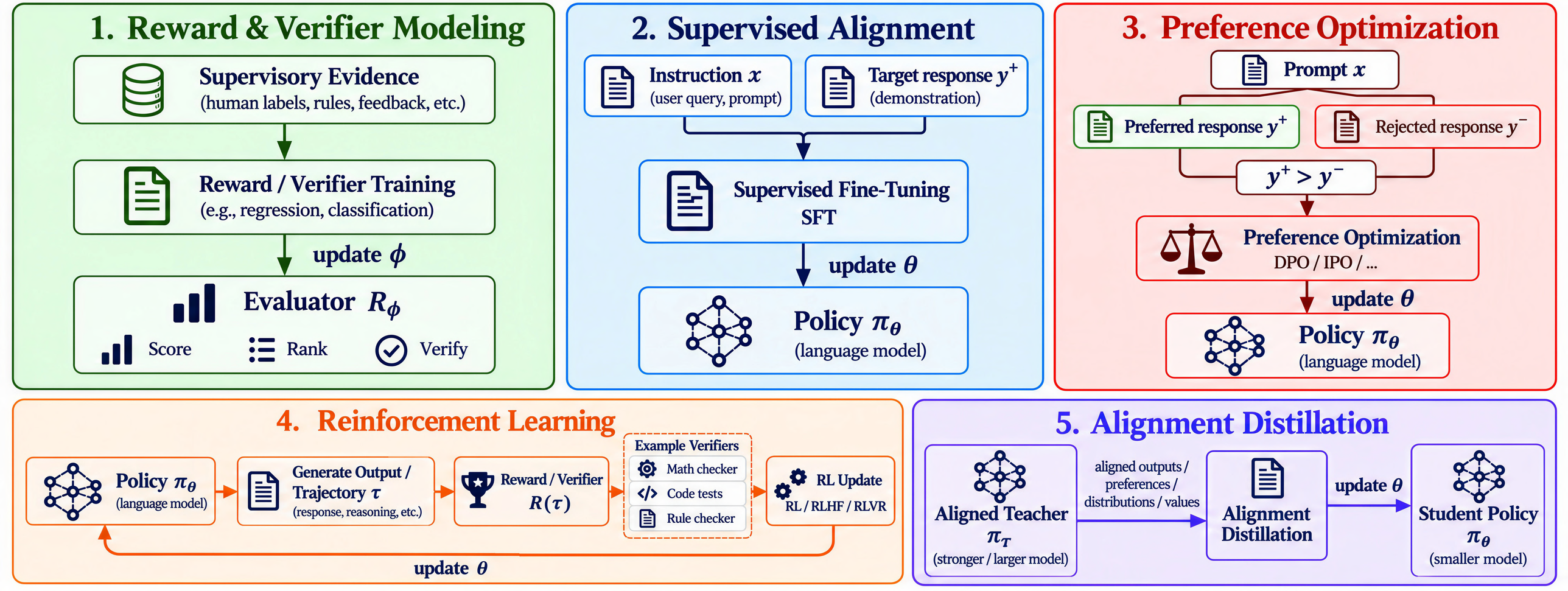

1. Reward & Verifier Modeling
Supervisory Evidence
(human labels, rules, feedback, etc.)
Reward / Verifier Training
(e.g., regression, classification)
update $\phi$
Evaluator $R_\phi$
Score
Rank
Verify
2. Supervised Alignment
Instruction $x$
(user query, prompt)
Target response $y^+$
(demonstration)
Supervised Fine-Tuning
SFT
update $\theta$
Policy $\pi_\theta$
(language model)
3. Preference Optimization
Prompt $x$
Preferred response $y^+$
Rejected response $y^-$
$y^+ > y^-$
Preference Optimization
DPO / IPO / ...
update $\theta$
Policy $\pi_\theta$
(language model)
4. Reinforcement Learning
Policy $\pi_\theta$
(language model)
Generate Output / Trajectory $\tau$
(response, reasoning, etc.)
Reward / Verifier
$R(\tau)$
Example Verifiers
Math checker
Code tests
Rule checker
RL Update
RL / RLHF / RLVR
update $\theta$
5. Alignment Distillation
Aligned Teacher
$\pi_T$
(stronger / larger model)
aligned outputs / preferences / distributions / values
Alignment Distillation
update $\theta$
Student Policy
$\pi_\theta$
(smaller model)